19-2-15

1) $y[n] = \sum_{k=-\infty}^{n-1} \alpha^{n-k} \{x[k-\beta_1] + \gamma x[k-\beta_2]\}$, $\alpha, \gamma \in \mathbb{R}$ $\beta_1, \beta_2 \in \mathbb{Z}$, $\beta_2 > \beta_1$

a) Si la entrada es una delta → $y[n] = h[n]$

$$h[n] = \sum_{k=-\infty}^{n-1} \alpha^{n-k} \{ \underbrace{\delta[k-\beta_1]}_{=1 \text{ para } k=\beta_1} + \gamma \underbrace{\delta[k-\beta_2]}_{=1 \text{ para } k=\beta_2} \}$$

= ~~δ ... k~~

$$h[n] = \begin{cases} Ⓐ\ \alpha^{n-\beta_1} & \text{si } n \le \beta_1 \\ Ⓑ\ \alpha^{n-\beta_1} + \gamma\alpha^{n-\beta_2} & \text{si } n \le \beta_2 \\ 0 & \text{si } n < \beta_1 \end{cases}$$

$\rightarrow [\alpha^{n-\beta_1} \cdot u[n-\beta_1]]$

$\Rightarrow [\alpha^{n-\beta_1} + \gamma\alpha^{n-\beta_2}]$

$[\alpha^{n-\beta_1} u[n-\beta_1] + \gamma\alpha^{n-\beta_2} u[n-\beta_2]]$

$m = n - \beta_1$

$\alpha^m u[m] = \dfrac{1}{1-\alpha z^{-1}}$ $\Rightarrow$

Ⓐ: $\dfrac{1}{1-\alpha z^{-1}}$ $n \le \beta_1$

Ⓑ: $\dfrac{1}{1-\alpha z^{-1}} + \dfrac{\gamma}{1-\alpha z^{-1}} = \dfrac{1+\gamma}{1-\alpha z^{-1}}$ $n \le \beta_2$

Ⓐ: Polo $1 - \frac{\alpha}{z} = 0 \Rightarrow 1 = \frac{\alpha}{z} \Rightarrow \boxed{z = \alpha}$

Ⓑ: Polo : $\boxed{z = \alpha}$ n ~~[illegible]~~

$|\alpha| \le 1$

$\alpha^{n-\beta_1} \cdot u[n-\beta_1] \overset{z}{\Longrightarrow} z^{-\beta_1} \dfrac{1}{1-\alpha z^{-1}} = \boxed{\dfrac{z^{-\beta_1}}{1-\alpha z^{-1}}}$

$\gamma\alpha^{n-\beta_2} u[n-\beta_2] \overset{z}{\Longrightarrow} \gamma z^{-\beta_2} \dfrac{1}{1-\alpha z^{-1}} = \dfrac{\gamma z^{-\beta_2}}{1-\alpha z^{-1}}$

$$\boxed{H_z[z] = \frac{z^{-\beta_1} + \gamma z^{-\beta_2}}{1-\alpha z^{-1}}}$$

~~$1 - \frac{\alpha}{z}$~~

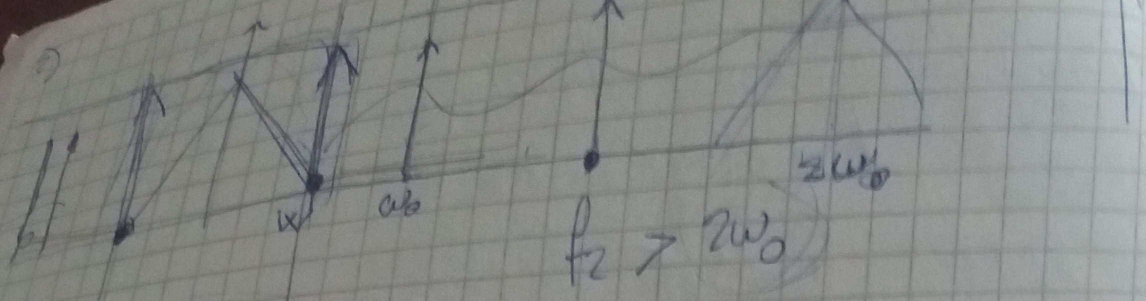
polo es α

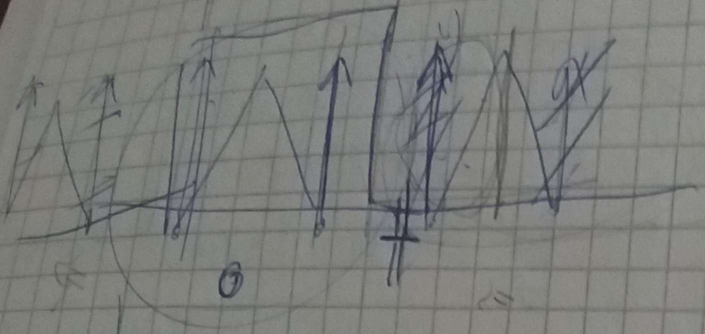
Ceros: $\dfrac{1}{z^{\beta_1}} + \dfrac{\gamma}{z^{\beta_2}} = 0 \Rightarrow \dfrac{1}{z^{\beta_1}} = \dfrac{-\gamma}{z^{\beta_2}} \Rightarrow \boxed{z^{\beta_2} = -\gamma z^{\beta_1}}$

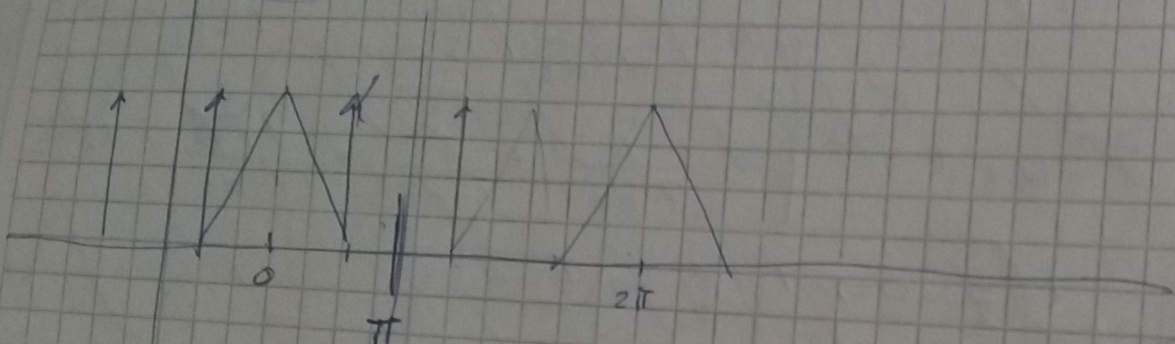
$$\boxed{z^{\beta_2} = -\gamma z^{\beta_1}} \Rightarrow z^{\beta_2} = -\gamma \text{ si } \beta_1 = 0$$

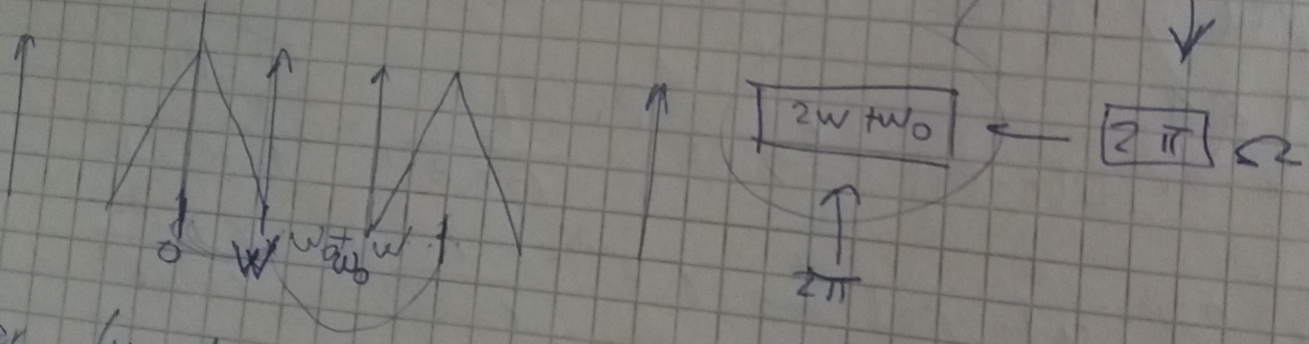
¿z = 0?

z = 0 ← cero

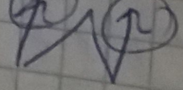




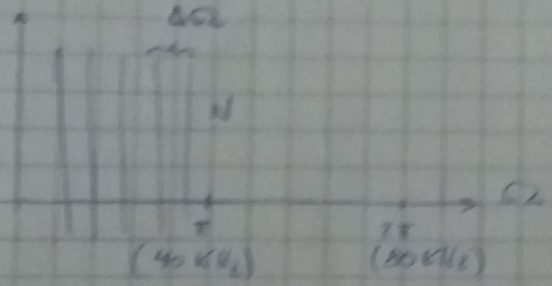




Setear (b) α para matar :



3a)

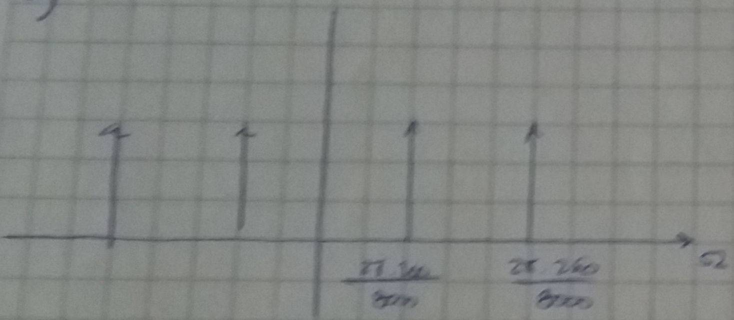


$\Delta\Omega = \frac{2\pi}{N} \rightarrow \Delta\omega = \Delta\Omega \cdot f_2 \rightarrow \Delta\omega = \frac{2\pi}{N} \cdot 80\ KHz$

Quiero $\Delta\Omega < 50\ Hz \rightarrow \cancel{2\pi} \cdot 50\ Hz > \frac{\cancel{2\pi}}{N} \cdot 80\ KHz$

$N > \frac{\cancel{2\pi} \cdot 8000\cancel{0}}{5\cancel{0}} \rightarrow N > 1600$

b)



Al utilizar N muestras se ventanea con una cuadrada por lo que el espectro viene dado por la conv de cada delta con una sinc().

Se requiere $\Delta\Omega > \frac{4\pi}{N}$

$\frac{2\pi\,250}{8000} - \frac{2\pi\,200}{8000} = \frac{4\pi}{N}$

$\frac{\cancel{2\pi}\,50}{8000} = \frac{4\pi}{N} \Rightarrow N = \frac{2\pi \cdot 8000}{50}$